UNIVERSITY ENTRANCE TEST FOR ADMISSION TO UNDER-GRADUATE PROGRAMMES – 2019

SERIES **D**

Booklet No: 113085

Time allowed : 2 hours 15 minutes for PCB & PCM and 3 hours for PCMB

This booklet contains 180 questions and 24 pages
Do not open this booklet until you are asked by the invigilator.

Instructions to be followed:

1. Immediately after opening the Question Booklet, ensure that it contains the number of pages and questions as depicted above. In case of any discrepancy, get the question booklet replaced.
2. There are 180 MCQs under Physics, Chemistry, Biology and Mathematics with 45 questions in each section. Students appearing under test stream PCB will attempt questions numbered between 1 to 135, those appearing under test stream PCM will attempt questions numbered between 1 to 90 and 136 to 180 and those appearing under test stream PCBM will attempt all the 180 questions. However, for Mathematics section (Q.No. 136 to 180), separate OMR shall be provided after submission of filled OMR of PCB by the student to the invigilation staff.
3. Use only blue/black pen for writing your particulars like Name, Roll No.; Series, Stream and Question Booklet No. on the OMR answer sheet at spaces provided. OMR answer sheets without mentioning Series or Question Booklet No. will not be evaluated.
4. Use only blue/black pen for writing your most appropriate responses on the OMR answer sheet.
5. Do not write anything else on OMR answer sheet. For rough work use the spaces provided on question booklet. Erasing, cutting or overwriting on the OMR sheet in answering circles will be deemed as incorrect answer and shall result in zero marks.
6. There shall be no negative marking for wrong answers.
7. The candidates should on demand show their admit cards to invigilators or any other authorized representative of the University.
8. No candidate will be allowed to leave the examination hall before half time except under special circumstances and permission by Centre Superintendent.
9. The candidates shall hand over the OMR answer sheet/s to invigilator before leaving the examination hall.
10. Use of electronic gadgets such as calculators, mobiles, Bluetooth devices, pagers, smart watches etc. are not allowed in the examination hall.
11. Any act of indiscipline/misconduct/use of unfair means shall result in forfeiture of candidate's right to sit in the examination. The decision of superintendent of the Centre in this regard shall be the final.
12. The cases pertaining to the use of unfair means will be governed by the rules and regulations of the University.

PHYSICS

Q1 Two points P and Q are maintained at the potential 10V and -4V respectively. The work done in moving 100 electrons from P to Q is:

A -9.60×10^{-17}J B 9.60×10^{-17}J
C -2.24×10^{-16}J D 2.24×10^{-16}J

3×10 10V q -4V Q -4V P 10V 100e

Q2 A hollow insulated conducting sphere is given a positive charge of 10 μC. What will be the electric field at the centre of the sphere, if its radius is 2 m?

A Zero B $5 \mu C m^{-2}$
C $20 \mu C m^{-2}$ D $32 \mu C m^{-2}$

Q3 What is the angle between the electric dipole moment and the electric field strength due to it on the equational line?

A 0° B 90°
C 180° D None of theses

Q4 Maximum lateral displacement of a ray of light incident on a slab of thickness "t" is:

A t B t/2
C t/3 D t/4

Q5 If a convex lens of focal length 75cm and a concave lens of focal length 50 cm are combined together, what will be their resulting power?

A +6.6D B 0.66D
C -6.6D D -0.66D

$f_1 = 75cm$ $f_2 = 50cm$ 125 $P = 1/f$

Q6 A current I flows in an infinitely long wire with cross Section in the form of a semicircular of radius R. The magnitude of the magnetic induction along its axis is:

A $\frac{\mu_0 I}{2\pi R}$ B $\frac{\mu_0 I}{4\pi R}$
C $\frac{\mu_0 I}{\pi^2 R}$ D $\frac{\mu_0 I}{2\pi^2 R}$

Q7 If number of turns, area and current through a coil are given by *n*, A and I respectively, then its magnetic moment is given by:

A nIA B n^2IA
C nIA^2 D $nI/\sqrt{A}$

Q8 Which of the following relation is not correct?

A $B = \mu_0(1 + H)$ B $B = \mu_0(1 + x_m)H$
C $I = \frac{B - \mu_0 H}{\mu_0}$ D $x_m = \mu - 1$

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$I = \frac{B - \mu_0 H}{\mu_0}$ $B = \mu_0(1+H)$

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Q9 For an electron in the second orbit of hydrogen, the angular momentum is:

A $2\pi h$ B πh
C h/π D $4/\pi$

Q10 de-Broglie wavelength of an electron in the first orbit of Bohr's hydrogen atom is:

A πr B $\pi r/2$
C $2\pi r$ D $2\pi r/3$

Q11 A certain radioactive sample has a half-life of 5 years. Thus for a nucleus on a sample of the element, probability of decay in 10 years is:

A 50% B 75%
C 60% D 100%

Q12 Which of the following radiations has the least wavelength?

A $\alpha - rays$ B $\beta - rays$
C $\gamma - rays$ D $X - rays$

Q13 Which of the following will deflect in electric field?

A $X - rays$ B $\gamma - rays$
C Cathode rays D Ultraviolet rays

Q14 Maximum velocity of photoelectron emitted is 4.8m/s. If e/m ratio of electron is 1.76 x 1011 (kg^{-1}), then stopping potential is given by:

A 5×10^{-10} V B 3×10^{-7} V
C 7×10^{-11} V D 2.5×10^{3} V

Q15 Which one of among the following shows particle nature of light?

A Polarization B Photoelectric effect
C Interference D Refraction

Q16 If the kinetic energy of the particle in increased to 16 times its previous value, the percentage change in the de-Broglie wavelength of the particle is:

A 60 B 50
C 25 D 75

Q17 A bullet fired into a fixed target loses half of its velocity after penetrating 3 cm. How much further it will penetrate before coming to rest?

A 1.5cm B 1.0cm
C 3.0cm D 2.0cm

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Q18 An automobile travelling with a speed of $60kmh^{-1}$ can brake to stop within a distance of 20 m. If the car is going twice as fast i.e $120kmh^{-1}$, the stopping distance will be:

A 29m B 40m

C 80m D 100m

Q19 Moment of inertia of a body does not depend upon its:

A Shape B Mass

C Angular velocity D Axis of rotation

Q20 The centre of mass of a solid cone along the line from the centre of the base to the vertex is at:

A One-forth of the height B One-third of the height

C One-fifth of the height D None of these

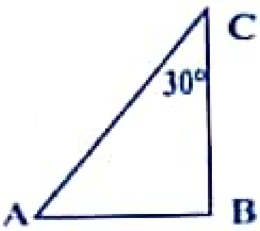
Q21 The angle of banking for a vertical speed of $10ms^{-1}$ for a radius of curvature 10m s $(G=10\ ms^{-2})$

$G = 10ms^{-2}$

A 45^0 B 30^0

C $\tan^{-1}\left[\frac{1}{2}\right]$ D 60^0

Q22 A body of mass M starts sliding down on the inclined plane where the critical angle ACB=30° as shown in figure. The coefficient of kinetic friction will be:



A $Mg/\sqrt{3}$ B $\sqrt{3}Mg$

C $\sqrt{3}$ D None of these

Q23 A monkey climbs up and another monkey climbs down a rope hanging from a tree with same uniform acceleration separately. If the respective masses of monkeys are in the ration 2:3, the common acceleration must be:

A g/5 B g/3

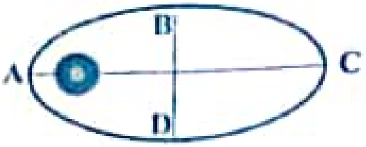
C g/2 D g

Q24 If the earth stops rotating suddenly, the value of g at a place other than poles would:

A Decrease B Remain Constant

C Increase D Increases or Decrease depending on the position of earth in the orbits round the sun.

Q25 In adjoining figure earth goes around the sun in elliptical orbit on which point the orbit speed is maximum:



A On A
B On B
C On C
D On D

Q26 If Kinetic Energy increases by 30%. Then momentum will increase by:

A 1.5%
B 9%
C 3%
D 2%

Q27 A mass of 0.5 kg moving with a speed of 1.5m/s on a horizontal smooth surface, collides with a nearly weightless spring of force constant k=50N/m. The maximum compression of the spring would be:

A 0.12m
B 1.5m
C 0.5m
D 0.15m

Q28 A pump is used to deliver water at a certain rate from a given pipe. To obtain "n" times water from the same pipe in the same time, by what factor the force of the motor should be increased?

A n times
B n^2 times
C n^3 times
D 1/n times

Q29 An ideal gas heat engine operates n Carnot cycle between 227°C and 127°C. It absorbs 6 x 104 cal of heat at higher temperature. The amount of heat converted into work is:

A 1.2×10^4 cal
B 2.4×10^4 cal
C 4.8×10^4 cal
D 6×10^4 cal

Q30 The given quantity of an ideal gas is at pressure P and absolute temperature T. The isothermal Bulk modulus of the gas is:

A $\frac{2}{3}P$
B P
C $\frac{3}{2}P$
D 2P

Q31 The moment of inertia of a uniform circular disc of radius R and mass M about an axis touching the disc at its diameter and normal to the disc is:

A MR^2
B $1/2MR^2$
C $2/5MR^2$
D $3/2MR^2$

Q32 Two bodies have their moments of inertia I and 2I respectively about their axis of rotation. If their kinetic energies of rotation are equal, their angular moments will be in the ratio of:

A 2:1
B 1:2
C $\sqrt{2}:1$
D $1:\sqrt{2}$

Q33 A cylinder is rolling down an inclined plane of inclination 60°. What is its acceleration?

A $g/\sqrt{3}$
B $g\sqrt{3}$
C $\sqrt{\frac{2}{3}}g$
D None of theses

Q34 A spring of force constant k is cut into two equal parts, which are then connected in parallel to each other. The force constant of the combination is:

A 4k
B 2k
C K
D k/2

Q35 A cylindrical resonance tube, open at both ends has a fundamental frequency "f" in air. If half of the length is dipped vertically in water, the fundamental frequency of the air column will be:

A $\frac{f}{2}$
B f
C $2f$
D $\frac{3f}{2}$

Q36 A child swinging on a swing in sitting position, stands up. Then the time period of the swing will:

A Increase
B Decrease
C Remains same
D Increase if child is long and decreases if child is short

Q37 Due to capillary action a liquid will rise in a tube if angle of contact is:

A Acute
B Obtuse
C 90°
D 180°

Deleted

Q38 If more air is pushed in a soap bubble, the pressure in it:

A Decrease
B Increase
C Remain same
D Becomes zero

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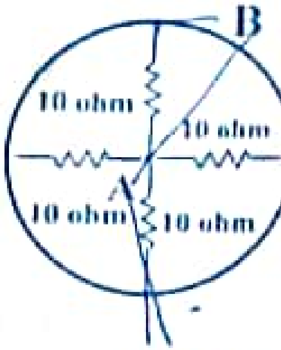
Q39 Find the equivalent resistance of the network shown in figure below between points A and B



A 1 ohm
C 2.5 ohm
B 2 ohm
D 3 ohm

Q40 A nichrome wire 1 m long and 1 mm² in cross-sectional area draws 4 A at 2 V. The resistivity of nichrome is:

A 1.0×10^{-7} ohm m
C 4.0×10^{-7} ohm m
B 2.0×10^{-7} ohm m
D 5.0×10^{-7} ohm m

Q41 Potentiometer measures the potential difference more accurately than a voltmeter, because:

A It does not draw current from external circuit
B It draws a heavy current from external circuit
C It has a wire of high resistance
D It has a wire of low resistance

Q42 The equation of state corresponding to 8kg of O_2 is:

A PV=RT
B PV=8 RT
C PV=RT/2
D PV=RT/4

Q43 The root mean square velocity of a gas molecule at 27°C is 1365 ms^{-1}. The gas is:

A O_2
B N_2
C He
D CO_2

Q44 Two non-reactive monoatomic ideal gases have their atomic masses in the ratio 2:3. The ratio of their partial pressures, when enclosed in a vessel kept at a constant temperature, is 4:3. The ratio of their densities is:

A 1:4
B 1:2
C 6:9
D 8:9

Q45 A step-down transformer is used to reduce 220 V supply to 11V. The primary draws 5 A of current, while the secondary 90 A. The efficiency of the transformer is:

A 20%
B 40%
C 70%
D 90%

CHEMISTRY

Q46 Which of the following is not an example of redox reaction?

A $CuO + H_2 \rightarrow Cu + H_2O$

B $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

C $2K + F_2 \rightarrow 2KF$

D $BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + 2HCl$

Q47 Which of the following compounds is used for water softening?

A $Na_6P_6O_{18}$

B Na_3PO_4

C Na_2HPO_4

D $Ca_3(PO_4)_2$

Q48 The element responsible for the neuromuscular function in the body is:

A Ca

B Fe

C K

D Na

Q49 Supercritical CO_2 is used as:

A Dry ice

B Fire fighting

C A solvent for extraction of organic compounds from natural resources

D A highly inert medium for carrying out various reactions

Q50 When formic acid is treated with conc. H_2SO_4 the gas evolved is:

A CO

B H_2S

C CO_2

D SO_3

CH

Q51 Based on lattice energy and other considerations, which one of the following alkali metal chlorides is expected to have highest melting point?

A LiCl

B NaCl

C KCl

D RbCl

Q52 Hybridization of Al in $AlCl_3$ (monomeric) and Al_2Cl_6 (dimeric) are :

A sp^2, sp^3

B sp^2, sp^2

C sp^3, sp^3

D sp^3, dsp^2

Q53 Slope of the plot between pv versus p at constant temperature is:

A 0

B ½

C 1

D $1/\sqrt{2}$

Q54 Which of the following is the uppermost region of the atmosphere:

A Exosphere

B Stratosphere

C Troposphere

D Ionosphere

Q55 The pH of normal rain water is

A 6.5

B 7.5

C 5.6

D 4.6

Q56 Green chemistry deals with :

A Study of Plant Physiology

B Study of extraction of natural products from plants

C Detailed study of reactions involved in synthesis of chlorophyll

D Utilization of existing knowledge base for reducing the chemical hazards along with developmental activities.

Q57 The crystal with metal deficiency defect is:

A KCl

B ZnO

C NaCl

D FeO

Q58 Rust is a mixture of:

A FeO and $Fe(OH)_2$

B Fe_2O_3 and $Fe(OH)_3$

C Fe_3O_4 and $Fe(OH)_3$

D None of these

Q59 Which of the following solution has highest equivalent conductance?

A 0.01MKCl

B 0.05 MKCl

C 0.02 MKCl

D 0.005 MKCl

Q60 The rate constant of a reaction is $175 L^2 mol^{-2} sec^{-1}$. What is the order of reaction?

A First

B Second

C Third

D Zero

Q61 Which of the following will have the highest coagulation power for As_2S_3 colloid?

A PO_4^{3-}

B Al^{3+}

C Na^+

D $[Fe(CN)_6]^{4-}$

Q62 Butter is a colloidal formed when

A Fat is dispersed in fat

B Fat is dispersed in water

C Water is dispersed in fat

D Suspension of casein in water

Deleted

Q63 The ore having two different metal atoms is:

A Haematite

B Galena

C Magnetite

D Copper pyrites

Q64 Which of the following oxidation state is common for all lanthanides?

A +2

B +5

C +3

D +4

Q65 In chromite ore, the oxidation number of Iron and Chromium are respectively

A +3, +2

B +3, +6

C +2, +6

D +2, +3

Q66 Vitamin B_{12} contains:

A Cobalt

B Magnesium

C Iron

D Copper

Q67 The molecule having smallest bond angle is:

A $AsCl_3$

B $SbCl_3$

C PCl_3

D NCl_3

Q68 **Normality of pure Sulphuric acid is:**

A 4
B 12
C 24
D 36

Q69 **The gas used in gas thermometer is:**

A He
B N_2
C O_2
D Ne

Q70 **The synthesis of alkyl fluorides is best accomplished by:**

A Finkelstein reaction
B Swarts reaction
C Free radial fluorination
D Sand Meyer's reaction

Q71 **Strongest acid among the following is:**

A o-methoxyphenol
B p-methoxyphenol
C m-methoxyphenol
D Phenol

Q72 **Acid that turns pink on exposing to air is:**

A Carbonic acid
B Oxalic acid
C Carbolic acid
D Citric acid

Q73 **Peptization refers to:** deleted

A Digestion of food
B Hydrolysis of proteins
C Breaking of dispersion into colloidal state
D Precipitation of solid from colloidal state.

Q74 **Number of structural isomers possible with an amine with molecular formula $C_4H_{11}N$ is:**

A 4
B 7
C 5
D 8

Q75 **Benzene diazonium Chloride on treatment with warm water gives:**

A Diphenyl ether
B p-Hydroxyazobenzene
C Phenol
D Chlorobenzene

Q76 **The base present in RNA but not in DNA is:**

A Guanine
B Cytosine
C Uracil
D Thymine

Q77 **Soft drinks and baby feeding bottles are generally made up of:**

A Polyester
B Polyurea
C Polystyrene
D Polyamide

Q78 **Barbutyric acid is used as:**

A An antipyretic
B An antiseptic
C An antibiotic
D A tranquilizer

Q79 **A measured temperature on Fahrenheit scale is 200°F, what will this reading be on Celsiu scale?**

A 40°C
B 94°C
C 93.3°C
D 30°C

Q80 **When the electron of a Hydrogen atom jumps from n=4 to n=1 state, the number of spectra lines emitted is:**

A 9
B 12
C 8
D 6

Q81 Which of the following has maximum spin?

A Electron
B Proton
C Neutron
D All have equal spin

Q82 The electronic configuration of Ce^{3+} is:

A $[Xe]4f^{13}$
B $[Kr]4f^{1}$
C $[Xe]4f^{1}$
D $[Kr]4d^{1}$

Q83 The best method for the separation of naphthalene and benzoic acid from their mixture is:

A Sublimation
B Chromatography
C Crystallisation
D Distillation

Q84 0.30g of an organic compound containing C, H and O on combustion yields 0.44g CO_2 and 0.18g H_2O. If one mole of compound weighs 60 then molecular formula of compound is:

A C_3H_8O
B $C_2H_4O_2$
C CH_2O
D C_4H_6O

Q85 Among the following compounds, the strongest acid is:

A C_2H_2
B C_6H_6
C C_2H_6
D CH_3OH

Q86 Propyne and Propene can be distinguished by:

A Con. H_2SO_4
B Br_2 in CCl_4
C Alk. $KMNO_4$
D $AgNO_3$ in NH_3

Q87 The state of a gas can be described by quoting the relationship between ....

A Pressure, volume, temperature
B Temperature, amount, pressure
C Amount, volume, temperature
D Pressure, volume, temperature, amount

Q88 Based on first law of thermodynamics, which of the following is correct?

A For an isochoric process, $\Delta U = -q$
B For an adiabatic process, $\Delta U = -w$
C For an isothermal process, $q = +w$
D For a cyclic process, $q = -w$

Q89 The Vapor Pressure of a liquid in a closed vessel

A Has a constant value depending only on the nature of the liquid.
B Keeps on increasing continuously as more and more liquid evaporates.
C Depends upon the amount of the liquid taken.
D Has a constant value at constant temperature.

Q90 Which one of the following ionic species has the greatest proton affinity to form stable compounds?

A I^-
B HS^-
C NH_2^-
D F^-

BIOLOGY

Q91 Ecotone is:
A A polluted area
B A zone of transition between two communities
C A zone of developing community
D The bottom of a lake

Q92 X linked recessive gene is
A Always expressed in male
B Always expressed in female
C Lethal
D Sub Lethal

Q93 Gel electrophoresis is used for:
A Separation of DNA fragments according to their size
B Isolation of DNA molecules
C Construction of recombinant DNA by joining with cloning vectors
D Cutting of DNA into fragments

Q94 Who invented DNA finger printing
A Alex Jeffreys
B Francois Jacob
C D. Ballimorr
D Hugo de Vries

Q95 The Lac Operon consists
A Four regulatory genes only
B One regulatory and three structural genes
C Two regulatory and two structural genes
D Three regulatory and three structural genes

Q96 The commonly used vectors for human genome sequencing are
A T – DNA
B T/A cloning vectors
C BAC and YAC
D Expression vectors

Q97 Breeding of crops with high level of minerals, vitamins and protein is called:
A Biomagnification
B Biofortification
C Bioluminiscence
D Micropropagation

Q98 PCR is required for:
A DNA amplification
B Protein synthesis
C DNA synthesis
D None of these

Q99 In Hardy Weinberg Law (equation) the frequency of heterozygous individuals is represented by
A p^2
B 2 pq
C q^2
D pq

Q100 A nucleoside differ from a nucleotide in not having
A Sugar
B Glucose
C Phosphate group
D Nitrogen base

Q101 Which of the following bacterium is used extensively as biopesticide:
A *Streptococcus lactis*
B *Bacillus thuringiensis*
C *Bacillus subtilis*
D *Lactobacilly acidophilus*

Q102 Aedes mosquito is a vector of:
A Malaria
B Filariasis
C Cholera
D Dengue

Q103 Botulism is caused due to contamination by :
A Streptococcus
B Escherichia coli
C Clostridium
D Pseudomonas

Q104 Association between roots of higher plants and fungi is called :
A Lichens
B Mycorrhiza
C Fern
D BGA

Q105 The Giant Redwood Tree (*Sequoia sempervirens*) is an :
A Pteridophyte
B Angiosperm
C Gymnosperm
D Bryophyte

Q106 Excretory organ in cockroach are :
A Malphigian tubules
B Nephredia
C Kidney
D Antennary glands

Q107 Name the phylum which exhibits sexual dimorphism
A Aschelminthes
B Arthropoda
C Platyhilminthes
D Annelida

Q108 Inflorescence in Cauliflower is :
A Corymb
B Catkin
C Umbel
D Compound Corymb

Q109 Placenta and Pericarp are both edible in :
A Tomato
B Banana
C Apple
D Potato

Q110 Nodulated roots are present in family
A Solanaceae
B Fabaceae
C Liliaceae
D None of these

Q111 Connecting link between fishes and amphibians is :
A Lung fish
B Sea horse
C Silver fish
D Star fish

Q112 Closed vascular bundles lack
A Ground tissue
B Pith
C Cambium
D Epidermis

Q113 Guttation occurs through :
A Stomata
B Hydathodes
C Cortex
D Both A and C

Q114 Proton Transport Theory of stomatal opening was proposed by :
A Levitt
B J.B. Sayre
C Priestley
D Steward

Q115 Which of the following sets of diseases is caused by bacteria:
A Typhoid and Small-pox
B Cholera and Tetanus
C Tetanus and Mumps
D Herps and Malaria

Q116 Kranz anatomy is found in leaves of :
A Sugarcane
B Maize
C Both A and B
D None of these

Q117 Which metal is constituent of chlorophyll ?
A Iron
B Zinc
C Magnesium
D Copper

Q118 Compensation point in plants is reached when
A Rate of photosynthesis is equal to rate of respiration
B Rate of photosynthesis is more than rate of respiration
C Rate of respiration is more than rate of photosynthesis
D None of these

Q119 The site of Krebs cycle is
A Mitochondrial matrix
B Cytoplasm
C Inter membrane space of mitochondria
D Ribosome

Q120 Growth in plants is measured by :
A Photometer
B Auxanometer
C Sphygmomanometer
D Clinostat

Q121 Food chain is always:
A Straight
B Curved
C Zig Zag
D Circular

D ✓

Q122 Wind pollination is common in
A Lilies
B Legumes
C Grasses
D Orchids

Q123 Part of Mango fruit eaten is :
A Mesocarp
B Epicarp
C Endocarp
D Tesla

Q124 Ribosome was discovered by
A George Palade
B Robert Brown
C Kolliker
D Carolus Linnaeus

Q125 The sequence of ecological succession occurring on sand is called:
A Lithosere
B Psammosere
C Xerosere
D Hydrosere

Q126 Which chemical is used to induce polyploidy:
A Nitrous Acid
B Colchicine
C IAA
D Cytokinin

Q127 T_i plasmid is obtained from:
A Azobacter
B Agrobacterium
C Yeast
D Rhizobium

Q128 Sugar present in milk is
A Sucrose
B Fructose
C Lactose
D Glucose

Q129 Distortion of tertiary structure of proteins under high temperature is called
A Renaturation
B Denaturation
C Titration
D Both A and B

Q130 The protective membrane of lung is called
A Pleuron
B Pericardium
C Meninges
D Peritoneum

Q131 Beri – Beri is caused due to deficiency of
A Vitamin A
B Vitamin B_1
C Vitamin B_2
D Vitamin C

Q132 Rate of heart beat is determined by
A S A node
B A V node
C Purkinje fibre
D Papillary muscle

Q133 Heart develops from
A Ectoderm
B Mesoderm
C Endoderm
D Both B and C

Q134 Copper ions released from copper releasing intra uterine devices (IUDs)
A Prevent ovulation
B Increase phagocytoses of sperms
C Suppress sperm motility
D Make uterus unsuitable for implantation

Q135 Ectopic pregnancies are referred to as :
A Pregnancy terminated due to hormonal imbalance
B Pregnancy with genetic abnormality
C Implantation of embryo at site other than uterus
D Implantation of defective embryo in the uterus

MATHEMATICS

Q136 In the expansion $(1+a)^{m+n}$, the cofficients of a^m and a^n are:

A Equal
B Not equal
C Do not say anything
D None of these

Q137 If mean of a Poisson distribution of a random variable x is 2, then the value of P(X>15) is:

A $\frac{3}{e^2}$
B $\frac{3}{e}$
C $1-\frac{3}{e}$
D $1-\frac{3}{e^2}$

Q138 A die is thrown three times

E: 4 appears on the third

F: 6 and 5 appears, respectively on first two tosses then $P\left(\frac{E}{F}\right)$ is:

A $\frac{1}{36}$
B $\frac{5}{6}$
C $\frac{5}{36}$
D $\frac{1}{6}$

Q139 For two events A and B, if P(A) and $P\left(\frac{A}{B}\right)=\frac{1}{4}$ and $P\left(\frac{B}{A}\right)=\frac{1}{2}$ then

A A and B are independent
B $P\left(\frac{A'}{B}\right)=\frac{3}{4}$
C $P\left(\frac{B'}{A'}\right)=\frac{1}{2}$
D All of the above

Q140 Two dice are thrown simultaneously. The probability of getting a pair of aces is:

A $\frac{1}{36}$
B $\frac{1}{3}$
C $\frac{1}{6}$
D None of these

Q141 The probability that a leap year will have 53 Fridays or 53 Saturdays is:

A $\frac{2}{7}$
B $\frac{3}{7}$
C $\frac{4}{7}$
D $\frac{1}{7}$

Q142 The median of distribution 83,54,78,64,90,59,67,72,70,73 is:

A 71
B 70
C 72
D 73

Q143 If $A = \{3, 5, 7, 9, 11\}$; $B = \{7, 9, 11, 13\}$; $C = \{11, 13, 15\}$ and $D = \{15, 17\}$ *then* $(A \cup D) \cap (B \cup C)$ *is equal to :*

A $\{5, 7, 9, 11, 15\}$ B $\{7, 9, 11, 15\}$
C $\{7, 9, 11, 13, 15\}$ D None of these

Q144 The domain of the function $f(x) = \dfrac{x^2+2x+1}{x^2-x-6}$ is :

A $R - \{3, -2\}$ B $R - \{-3, 2\}$
C $R - \{-3, -2\}$ D None of the above

Q145 The range of $f(x) = \dfrac{1}{1-2\cos x}$ is :

A $\left[\frac{1}{3}, 1\right]$ B $\left[-1, \frac{1}{3}\right]$
C $(-\infty, -1) \cup \left[\frac{1}{3}, \infty\right]$ D $\left[-\frac{1}{3}, 1\right]$

Q146 The expression $\cos\left(\dfrac{3\pi}{2}+x\right)\cos(2\pi+x)\left[\cot\left(\dfrac{3\pi}{2}-x\right)+\cot(2\pi+x)\right]$ is equal to :

A 1 B 0
C 2 D -1

Q147 The value of $\dfrac{1-\tan^2 15}{1+\tan^2 15}$ is :

A 1 B $\sqrt{3}$
C $\dfrac{\sqrt{3}}{2}$ D 2

Q148 The value of $\tan^{-1}\left\{2\cos\left(2\sin^{-1}\frac{1}{2}\right)\right\}$

A $\dfrac{\pi}{4}$ B $\dfrac{-\pi}{4}$
C $\dfrac{\pi}{3}$ D None of the above

Q149 By mathematical induction $\dfrac{1}{1.2.3}+\dfrac{1}{2.3.4}+\ldots\ldots+\dfrac{1}{n(n+1)(n+2)}$ is equal to

A $\dfrac{n(n+1)}{4(n+2)(n+3)}$ B $\dfrac{n(n+3)}{4(n+1)(n+2)}$
C $\dfrac{n(n+2)}{4(n+1)(n+3)}$ D None of the above

Q150 The term independent of x in $\left(3x - \frac{2}{x^2}\right)^{15}$ is :

A $-3003(3^{10})(2^5)$ B $-3003(3^{10})(2^4)$

C $3003(3^{10})(2^5)$ D None of these

A=XB

Q151 If $3\begin{bmatrix} x & y \\ z & w \end{bmatrix} = \begin{bmatrix} x & 6 \\ -1 & 2w \end{bmatrix} + \begin{bmatrix} 4 & x+y \\ z+w & 3 \end{bmatrix}$ then the value of z is:

A 1 B 2

C 3 D 4

Q152 A 2 X 2 matrix A = $[a_{ij}]$ whose elements are given 6y $a_{ij} = \frac{(i+2j)^2}{2}$ is :

A $\begin{bmatrix} 1 & 1/2 \\ 2 & 1 \end{bmatrix}$ B $\begin{bmatrix} 9/2 & 25/2 \\ 8 & 8 \end{bmatrix}$

C $\begin{bmatrix} 8 & 9/2 \\ 25/2 & 8 \end{bmatrix}$ D None of the above

Q153 If $|A| = \begin{vmatrix} 3 & -1 & -2 \\ 0 & 0 & -1 \\ 3 & -5 & 0 \end{vmatrix}$ then the value is :

A -5 B -7

C -10 D -12

Q154 In an A.P, if the p^{th} term is $1/q$ & q^{th} term is $1/p$ then the sum of first pq term is:

A (pq+1) B $\frac{1}{2}(pq+1)$

C $\frac{1}{2}(pq-1)$ D None of the above

Q155 The slope of the line passing through points (3, – 2) and (3, 4) is:

A Undefined B 0

C 1 D 6

Q156 The equation of perpendicular bisector of the line joining the point A(2, 3) and B(6, – 5) is:

A $x-2y-6=0$ B $3x+y-6=0$

C $2x-2y+6=0$ D None of the above

Q157 For specifying a straight line, how many geometrical parameters should be known:

A 1 B 2

C 3 D 4

Q158 If the equation of parabola is $x^2 = 6y$, then the co-ordinates of focus and axis of the parabola are:

A $\left(0, \frac{3}{2}\right)$, $y-axis$
B $\left(0, \frac{-3}{2}\right)$, $y-axis$
C $\left(0, \frac{3}{2}\right)$, $x-axis$
D None of the above

Q159 The equation of ellipse with centre of origin, the length of major axis is 12 and one focus of (4, 0) is:

A $\frac{x^2}{16} + \frac{y^2}{9} = 1$
B $\frac{x^2}{36} + \frac{y^2}{20} = 1$
C $\frac{x^2}{9} + \frac{y^2}{9} = 1$
D None of the above

Q160 The equation of Hyperbola, whose Vertices are (0, ± 7) and e = $4/3$ is :

A $7y^2 + 8x^2 = 343$
B $7x^2 + 3y^2 = 121$
C $3y^2 + 8x^2 = 211$
D $7y^2 - 9x^2 = 343$

Q161 $\lim_{x \to 1} \frac{1 + x^{1/3}}{1 + x^{1/5}}$ is equal to:

A $3/5$
B $-5/3$
C $5/3$
D $-3/5$

Q162 If $y = \sin^{-1}\left[\frac{\log x^2}{1 + (\log x)^2}\right]$ then $\frac{dy}{dx}$ is equal to:

A $\frac{2}{1 + \log x}$
B $\frac{2}{x[1 + \log x]}$
C $\frac{2}{x[1 + (\log x)^2]}$
D $\frac{2}{1 + (\log x)^2}$

Q163 The angle of intersection of the curves $y = 2\sin^2 x$ and $y = \cos 2x$ at $x = \frac{\pi}{6}$ is :

A $\frac{\pi}{3}$
B $\frac{2\pi}{3}$
C both A & B
D None of the above

Q164 $\int \frac{dx}{\sqrt{9-25x^2}}$ is equal to:

A $\sin^{-1}\left(\frac{5x}{3}\right)+c$

B $\frac{3}{2}\sin^{-1}\left(\frac{5x}{3}\right)+c$

C $\frac{1}{5}\sin^{-1}\left(\frac{5x}{3}\right)+c$

D None of the above

Q165 $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x}\,X\sqrt{\cos x}}\,dx$ is equal to:

A 0

B $\frac{\pi}{4}$

C $\frac{\pi}{2}$

D None of the above

Q166 The solution of $x\log x\frac{dy}{dx}+y=\frac{2}{x}\log x$ is:

A $y\log x=\frac{1}{x}(1+\log x)+c$

B $y\log x=\frac{2}{x}(1+\log x)+c$

C $y\log x=2x(1+\log x)+c$

D $y\log x=\frac{-2}{x}(1+\log x)+c$

Q167 If $|a|=3$ $|b|=1$, $|c|=4$ and $a+b+c=0$ then the value of $a.b+b.c+c.a$ is :

A 13

B 26

C −26

D −13

Q168 If a and b are two collinear vectors, then which of the following are in correct?

A b= λ a, for some scalar λ

B $\hat{a}=\pm b$

C a and b are proportional

D Both the vectors a and b have same direction but different magnitude.

Q169 The points A (1, 2, 7), B (2, 6, 3) and C (3, 10, -1) are :

A Collinear

B Coplanar

C Non Collinear

D None of the above

Q170 If $a+b+c=0$ and $|a|=\sqrt{37}$, $b=3$, $|c|=4$, then the angle between b and c is:

A 30°

B 45°

C 60°

D 90°

Q171 The direction cosines of a line which makes equal angles with the co-ordinate axis are:

A $<\frac{1}{3},\frac{1}{3},\frac{1}{3}>$

B $<\frac{-1}{3},\frac{-1}{3},\frac{-1}{3}>$

C $<\pm\frac{1}{\sqrt{3}},\pm\frac{1}{\sqrt{3}},\pm\frac{1}{\sqrt{3}}>$

D $<\pm\frac{1}{3},\pm\frac{1}{3},\pm\frac{1}{3}>$

Q172 The angle between the pair of lines $r = 3\hat{i} + 3\hat{j} - 4\hat{k} + \lambda(\hat{i} + 2\hat{j} + 2\hat{k})$ and $r = 5\hat{i} - 2\hat{k} + 4(3\hat{i} + 2\hat{j} + 6\hat{k})$ is :

A 0 B $\frac{\pi}{2}$
C π D None of these

Q173 The maximum value of $z = 4x + 2y$ subject to the constraints $2x + 3y = 18, x + y \geq 10, x, y \geq 0$ is :

A 36 B 40
C 20 D None of these

Q174 The objective function $z = 4x + 3y$ can be maximized subjected to the constraints $3x + 4y \leq 24, 8x + 6y \leq 48, x \leq 5, y \leq 6, x, y \geq 0$

A At only one point B At two points only
C At an infinite number of points D None of these

Q175 The region represented by the inequation system $x, y \geq 0, y \leq 6, x + y \leq 3$ is:

A Unbounded in first quadrant B Unbounded in first and second quadrant
C Bounded in first quadrant D None of these

Q176 If the constraints in a linear programming problem are changed:

A The problem is to be re-evaluated B Solution is not defined
C The objective function has to be modified D The change in constraints is ignored

Q177 If A and B are two events such that $P(A) = \frac{4}{5}$ and $P(A \cap B) = \frac{7}{10}$ then $P\left(\frac{B}{A}\right) =$

A $\frac{1}{10}$ B $\frac{1}{8}$
C $\frac{7}{8}$ D $\frac{17}{20}$

Q178 The coefficient of x^5 in $(1+2x)^6 (1-x)^7$ is:

A -171 B 171
C 172 D -172

Q179 The middle term in the expansion $\left(x^2 + \frac{1}{x^2} + 2\right)^n$

A $\frac{2n!}{n!n!}$ B $\frac{2n!}{n!}$
C $\frac{2n!}{(n-1)!}$ D None of these

Q180 Find the coefficient of x^9 in $(1 + 3x + 3x^2 + x^3)^{15}$ is:

A $\frac{45!}{46!9!}$ B $\frac{45!}{46!8!}$
C $-\frac{45!}{46!9!}$ D $\frac{45!}{46!7!}$